

Online Retail Sales & Revenue Analysis

Dataset: UCI Online Retail | Period: December 2010 – December 2011 | Raw rows: 541,909 | Clean rows: 536,638

Introduction

This report summarises the exploratory data analysis (EDA) conducted on the UCI Online Retail dataset, a real-world transactional dataset from a UK-based gift wholesaler. The analysis focuses on revenue and sales trends across time, geography, and product lines. Data was cleaned and prepared before analysis using Python (pandas) in Google Colab. Visualizations were produced using matplotlib and seaborn.

Data Cleaning Challenges

The following issues were identified and addressed during the cleaning phase:

Category	Issue Found	Action Taken	Rows Affected	Status
Missing values	135,080 missing CustomerIDs	Retained — guest checkouts have valid revenue data	135,080	Kept
Missing values	1,454 missing Descriptions	Filled using StockCode lookup from existing rows	1,454	Filled
Duplicates	5,226 rows flagged	Investigated — found to be legitimate invoice line items, not true duplicates. No rows removed.	5,226	No action
Invalid entries	9,288 cancellation rows (InvoiceNo starting with C)	Removed — cancellations are not real sales	9,288	Removed
Invalid entries	Rows with Quantity or UnitPrice of 0 or below	Removed — not valid sales transactions	~10,000	Removed
Standardization	InvoiceDate as string, CustomerID as float	InvoiceDate parsed to datetime, CustomerID converted to integer	All rows	Applied
Feature engineering	No Revenue, YearMonth, DayOfWeek, Hour or CustomerType columns	Derived columns created from existing fields	All rows	Applied

Key EDA Findings

Revenue & Sales Trends

- £10.6M Total revenue over the 13-month period was
- Revenue followed a clear seasonal pattern, peaking in November 2011 at approximately £1.5M driven by Black Friday and pre-Christmas demand.

- Average order value was stable at £430–£580 per month, spiking to £780 in December.

Geography

- The United Kingdom dominates revenue, consistent with the business being a UK-based wholesaler.
- Top international markets are the Netherlands, EIRE (Ireland), Germany, and France.

Products

- DOTCOM POSTAGE is the top product by revenue but is likely a shipping charge, not a real product.
- PAPER CRAFT, LITTLE BIRDIE leads both revenue and quantity sold (~80,000 units) making it a high-volume, high-value product.
- REGENCY CAKESTAND 3 TIER is a premium item with high revenue but lower volume.

Operations & Time Patterns

- The business operates Monday to Friday only — no transactions were recorded on Saturdays, consistent with a B2B wholesale model.
- Thursday is the busiest day for orders. Peak ordering time is 12 noon.
- The weekly revenue heatmap confirms a significant spike on a Friday in late November (Black Friday).

Customer Type

- Registered customers account for ~£8.9M (84%) of total revenue.
- Guest checkouts contribute ~£1.7M (16%) — a meaningful share that should not be ignored.

Top Insights

- Q4 seasonality is the single biggest revenue driver — businesses should prepare inventory and staffing for September to November.
- Saturday inactivity confirms a B2B-only customer base — there is no weekend retail demand to capture.
- The Netherlands and EIRE are the strongest international markets and represent the best candidates for targeted international growth.
- Guest checkout revenue (16%) is significant — converting guests to registered accounts could meaningfully increase long-term revenue.
- DOTCOM POSTAGE appearing as the top product by revenue is a data quality flag — shipping charges should be excluded from product-level analysis.
- Noon is peak ordering time — promotional emails or flash deals timed around midday could maximise engagement.